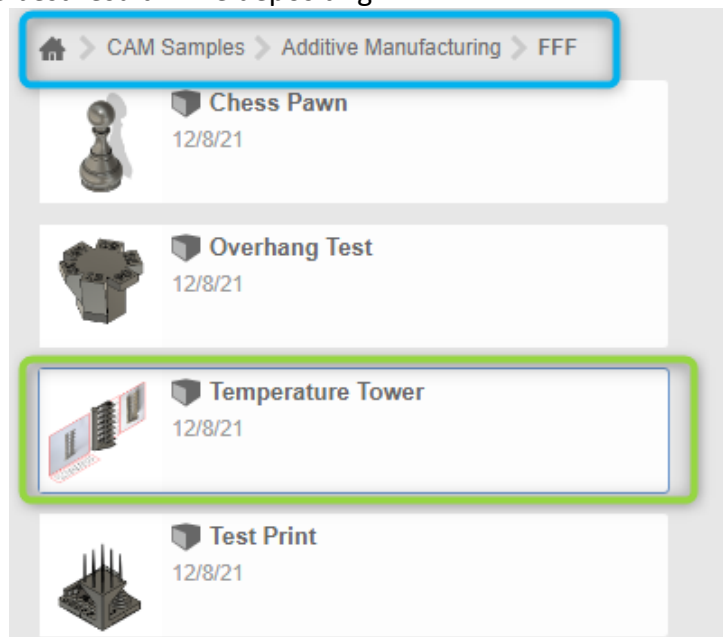


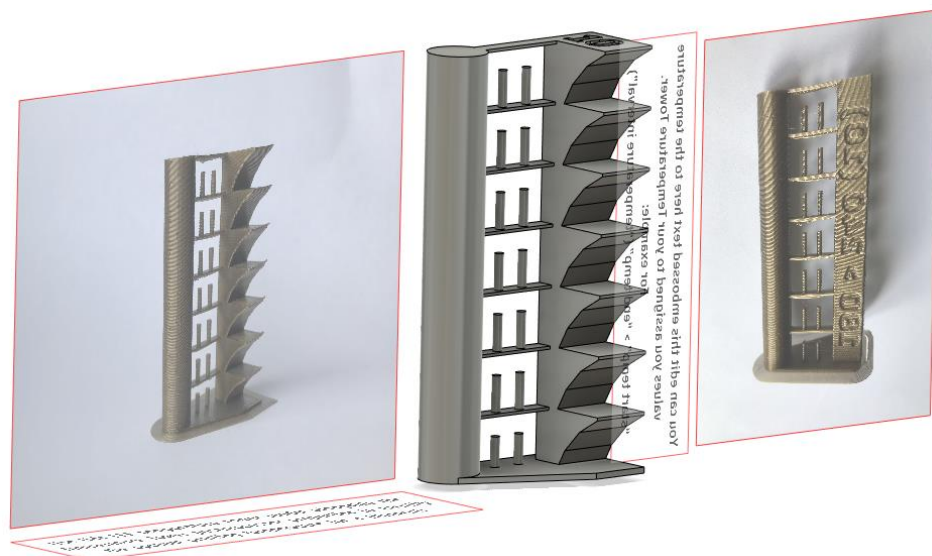


Using the Temperature tower functionality

If you look at the data panel in Fusion360, you can have access to a model, to use the temperature tower test. It's used for determining the optimal temperature, achieving the best result while depositing.



You can open this model and then generate the toolpath, after selecting the relevant print settings for the material, filament diameter, nozzle diameter combination. The model is this one :





So, we have seven steps which are 10mm high each, making a 70mm tower in total. This information will be useful when defining the parameters later.

Once the toolpath is generated, we can post process using either the legacy post dialog, or the newest Nc-program.

When using the Nc-program, the dialog look like this :

NC Program: NCProgram1

Settings

Machine and post

Use machine configuration ☒

Machine eMotion Tech Strateo3D I ...

Post Strateo3D DUAL600 ...

Use cascading post ☐

Program

Name/number 1234

Comment My Temp Tower test

Output folder d:\temp\

Post to Fusion Team ☐

NC extension .gcode

Unit Document unit

Open NC file in editor ☒

Post properties

Temperature Tower

Trigger Disabled

Trigger Value Disabled

Temperature Interval by Layer

Start Temperature 190

General

Built-in

The red area is the selected machine configuration, used for creating the toolpath.

The green area is the selected post.

Then the blue area is the post properties form.

Let's detail these properties for the temperature tower function.

Post properties

Temperature Tower

Trigger by Height

Trigger Value 10

Temperature Interval 5

Start Temperature 190

First one *Trigger*, is for selecting a temperature change, either from a layer count, or the z height.

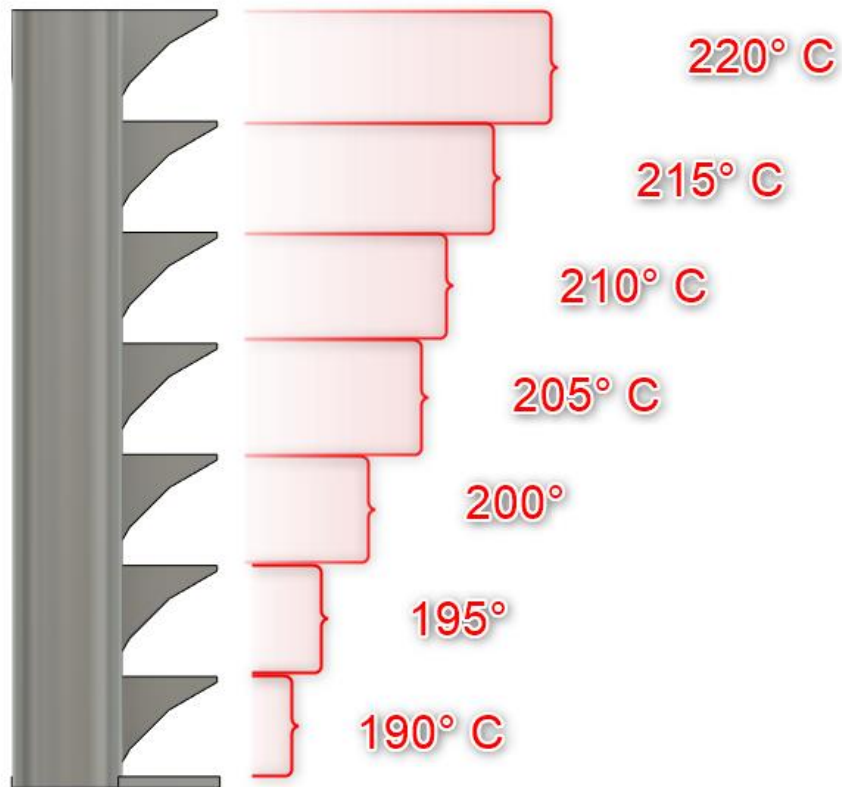
The *Trigger Value* will define either the number of layers, or the z height increment.



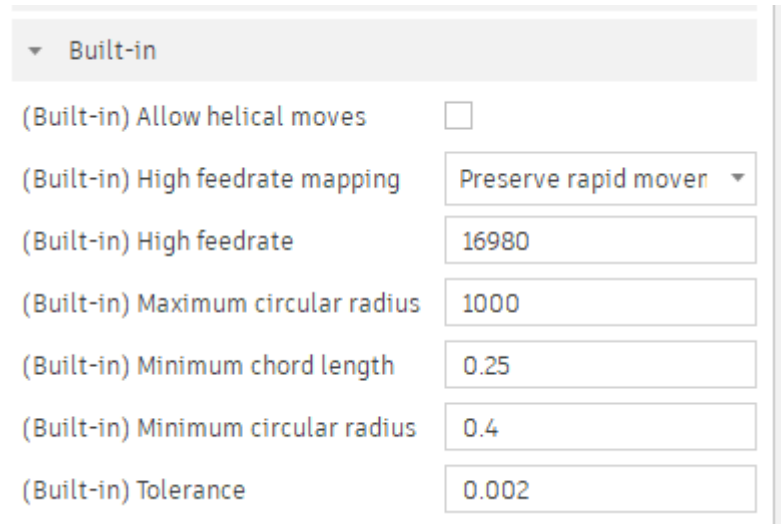
The *Start Temperature* is forcing the initial temperature for the print, overriding the one in the print settings.

The *Temperature Interval* give the increase of temperature at each triggered step.

Based on the model definition, and the current parameters, the result would be :



Usage of the Build-in post properties



Built-in	
(Built-in) Allow helical moves	☐
(Built-in) High feedrate mapping	Preserve rapid mover ▼
(Built-in) High feedrate	16980
(Built-in) Maximum circular radius	1000
(Built-in) Minimum chord length	0.25
(Built-in) Minimum circular radius	0.4
(Built-in) Tolerance	0.002

These properties define some capabilities useable by the post and settings related to the arc interpolation.

Let's group the properties by related functionality.

The high feed rate handling

The first property *High feedrate mapping* is inherited from the milling/turning post, and it has no effect for an additive fused filament fabrication post processor.

The *High feedrate* value define the highest feed possible for the printer, expressed in millimeter per minute.

The arc interpolation settings

The switch *Allow helical moves* have no action in an additive post, as we only support planar arcs. (The post is preventing any arcs other than in the XY plane)

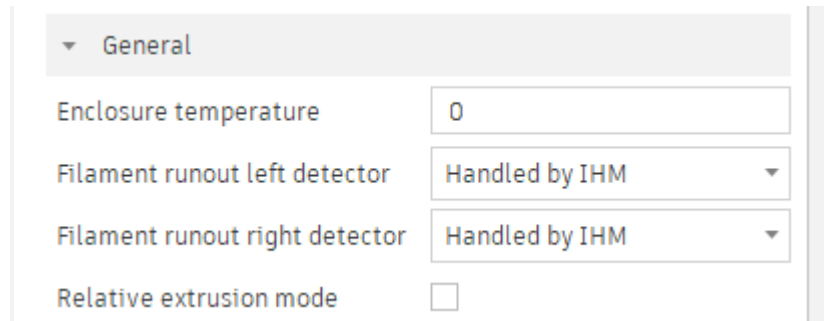
The value *Maximum circular radius* defines the biggest arc that can be interpolated. Bigger arcs will be linearized by the post engine.

The value *Minimum circular radius* defines the smallest arc, smaller will be linearized.

The *Minimum chord length* defines the smallest chord length before an arc will be linearized.

The *Tolerance* value will be used when linearizing the arcs. The smaller the value, the more segments we get.

The printer specific post properties



▼ General	
Enclosure temperature	0
Filament runout left detector	Handled by IHM ▼
Filament runout right detector	Handled by IHM ▼
Relative extrusion mode	☐

For the Starteo3D, we added a property to set the temperature to be maintained inside the enclosure. The valid values are between 0 and 60° C. If the value is 0, we do not output the M141 S command.

The list box Filament runout left detector allow to either enable, disable, or let the printer IHM define the state of the filament detector.



- Disabled
- Enabled
- Handled by IHM

We have the same capability for the right filament detector.

In the last case (IHM), no code is output in the g-code, otherwise we output an M405 or M406 depending on the choice made.

The Boolean value *Relative extrusion mode*, switch the extrusion from absolute (M82) to incremental (M83) mode. The E value is then outputted absolute or relative.